The fast food industry represents one of the most dynamic sectors in the stock market, with companies ranging from global giants like McDonald's to innovative newcomers like Luckin Coffee. In this project, you'll create an interactive line chart to visualize historical stock prices for 10 major fast-food companies. Building interactive dashboards is a critical skill in modern business, as these tools help analysts, portfolio managers, and executives make informed, data-driven decisions.

Dataset Summary

Your primary dataset, `companies.csv`, contains historical stock market data from Yahoo Finance. This dataset captures daily trading activity from major players across the fast food sector. It includes stock data from these industry leaders:

Ticker	Company	Description
BRK-A	Berkshire Hathaway Inc.	Financial conglomerate with major fast food investments
DNUT	Krispy Kreme, Inc.	Specialty donut and coffee retailer
DPZ	Domino's Pizza, Inc.	Global pizza delivery leader
LKNCY	Luckin Coffee Inc.	Chinese coffee chain competitor
MCD	McDonald's Corporation	World's largest fast food restaurant chain
PZZA	Papa John's International	Pizza delivery and takeout specialist
QSR	Restaurant Brands International	Parent of Burger King, Tim Hortons, and Popeyes
SBUX	Starbucks Corporation	Global coffeehouse chain
WEN	The Wendy's Company	Premium burger restaurant chain
YUM	Yum! Brands, Inc.	Parent of KFC, Taco Bell, and Pizza Hut

Note: A ticker is a unique stock symbol used to identify companies on exchanges (e.g., "MCD" for McDonald's, "SBUX" for Starbucks).

Data Structure

`companies.csv`

Column	Description
`date`	Trading date
`open`	Opening price for the trading session
`high`	Highest price during the day
`low`	Lowest price during the day

Column	Description
`close`	Closing price at market close
`adj_close`	Price adjusted for dividends and stock splits
`volume`	Number of shares traded
`company_ticker`	Company stock symbol identifier

```
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go

# Read the CSV file
companies = pd.read_csv('companies.csv')

# Ensure 'date' is in datetime format
companies['date'] = pd.to_datetime(companies['date'])

companies.head()
```

	date	open	high	low	close	adj_close	
0	2019-01-02T00:00:00.000	302000	306255	301880	304057	304057	4
1	2019-01-02T00:00:00.000	39.6399993896	40.6599998474	38.9199981689	40.1300010681	36.1643295288	3702
2	2019-01-02T00:00:00.000	245.5899963379	245.5899963379	240.2400054932	243.3000030518	228.5730743408	4447
3	2019-01-02T00:00:00.000	51.6100006104	52.0600013733	50.7799987793	51.4199981689	42.6107559204	19391
4	2019-01-02T00:00:00.000	91.0899963379	91.5400009155	90.3499984741	91.4400024414	82.3424835205	17434

Rows: 5 Expand

```python
# Start coding here. Use as many cells as you like!
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go

# Read the CSV file
companies = pd.read_csv('companies.csv')

# Ensure 'date' is in datetime format
companies['date'] = pd.to_datetime(companies['date'])

# --- Set up the list of company tickers, column labels, and default selections for the plot ---
tickers = ["BRK-A", "DNUT", "DPZ", "LKNCY", "MCD",
           "PZZA", "QSR", "SBUX", "WEN", "YUM"]

# OPTIONAL: friendly labels for cleaner dropdown text
col_map = {
    "open":  "Open",  "high": "High", "low": "Low",
    "close": "Close", "adj_close": "Adj Close", "volume": "Volume"
}

default_tkr = "DPZ"
default_col = "close"

# --- Build one trace per company for the default column ---
fig = go.Figure()
for t in tickers:
    sub = companies[companies["company_ticker"] == t]
    trace = px.line(sub, x="date", y=default_col).data[0]
    trace.visible = (t == default_tkr)  # OPTIONAL: default company only → cleaner first view
    fig.add_trace(trace)

# --- Dropdown for company selection; updates visible trace only ---
company_buttons = []
for i, t in enumerate(tickers):
    vis = [False] * len(tickers); vis[i] = True
    company_buttons.append({
        'label': t,
        'method': "update",
        'args': [
            {'visible': vis},
            {'title': f"{t} - {col_map[default_col]}"} # OPTIONAL: a dynamic title
        ]
    })

# --- Dropdown for column selection; updates y-data for all traces (not shown in course) ---
column_buttons = []
for c_key, c_lbl in col_map.items():
    y_lists = [companies[companies["company_ticker"] == tt][c_key]
               for tt in tickers]
    column_buttons.append({
        'label': c_lbl,
        'method': "update",
        'args': [
            {'y': y_lists},  # Updates y-data for each trace; this pattern not shown in slides
            {'title': f"{default_tkr} - {c_lbl}"}
        ]
    })

# --- Range-selector buttons for time filtering ---
range_buttons = [
    {'count': 1, 'step': "month", 'stepmode': "backward", 'label': "1M"},
    {'count': 6, 'step': "month", 'stepmode': "backward", 'label': "6M"},
    {'count': 1, 'step': "year",  'stepmode': "backward", 'label': "1Y"},
    {'count': 3, 'step': "year",  'stepmode': "backward", 'label': "3Y"}
]

# --- Update layout with two dropdowns and time buttons ---
fig.update_layout({
    'title': f"{default_tkr} - {col_map[default_col]}",
    'xaxis': {
        'rangeselector': {'buttons': range_buttons},
```

```
        'rangeslider':   {'visible': False},
        'type': 'date'
    },
    'updatemenus': [
        {'type': "dropdown", 'direction': 'down',
         'x': 1.12, 'y': 1.00, 'showactive': True,
         'buttons': company_buttons,
         'active': tickers.index(default_tkr)},
        {'type': "dropdown", 'direction': 'down',
         'x': 1.12, 'y': 0.6, 'showactive': True,
         'buttons': column_buttons,
         'active': list(col_map).index(default_col)}
    ],
    'hovermode': "x unified"  # OPTIONAL: unified tooltip → less clutter
})

fig.show()
```